If $F_1 < 0$, the lower 4 balls separate. For these 4, equation (8) gives:

$$5\tan\beta - \cot\alpha < 0 \tag{9}$$

Using (2) and (5), (9) becomes:

$$\frac{5\sqrt{2}a}{\sqrt{R^2 - 2Ra - a^2}} < \frac{\sqrt{2}a}{\sqrt{b^2 + 2ab - a^2}} \tag{10}$$

Solving the inequality gives:

$$R > \sqrt{25b^2 + 50ab - 23a^2} + a \tag{11}$$

For bowl radius R as above, the lower balls separate.

For (11) to have a solution:

$$25b^2 + 50ab - 23a^2 \geqslant 0 \tag{12}$$

i.e.

$$b \geqslant \left(\frac{4}{5}\sqrt{3} - 1\right)a \tag{13}$$

The least distance from O is $(\sqrt{2} - 1)a$. A ball on the 4 lower balls must therefore satisfy

$$b > (\sqrt{2} - 1)a \tag{14}$$

Since $\sqrt{2} > 4\sqrt{3}/5$, a 5th ball resting on 4 balls makes (12) hold automatically.

2.

Before launch the probe is part of the Earth satellite, sharing its elliptical orbit. Major and minor radii are a, b; center O is c from focus F. Fig. 20-2-1 and geometry give

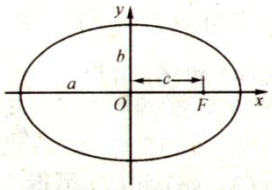


Fig. 20-2-1

$$c^2 = a^2 - b^2 \tag{1}$$

Before launch it has orbital energy. Speed at apogee is v_1; its energy before launch is

$$E_1 = \frac{1}{2}mv_1^2 - \frac{Gm_E m}{a + c} \tag{2}$$

Gravity always acts through Earth's center, giving zero torque. Thus angular momentum is conserved. At perigee speed v_2, conservation gives:

$$mv_1(a + c) = mv_2(a - c) \tag{3}$$

Gravity is conservative: orbital energy is conserved. At apogee and perigee:

$$\frac{1}{2}mv_1^2 - \frac{Gm_E m}{a + c} = \frac{1}{2}mv_2^2 - \frac{Gm_E m}{a - c} \tag{4}$$

Solving (3) and (4):

$$v_1^2 = \frac{Gm_E}{a}\frac{a - c}{a + c} \tag{5}$$

Substituting (5) into (2):

$$E_1 = \frac{Gm_E m}{2a} \tag{6}$$

The ejected probe moves on a hyperbola whose vertex is the old elliptical vertex (Fig. 20-2-2). Angular momentum and mechanical energy are conserved. Let v_0 be the ejection speed. At the vertex, the angular momentum is $mv_0(a + c)$, and Earth-center distance is $a + c$. At infinity velocity follows the asymptote. Its speed is v. Let d be the distance from Earth's center to that line.

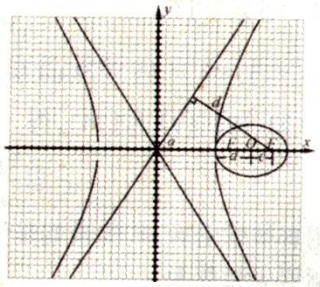


Fig. 20-2-2